



High frequency (RF)
 $RF \& LO = IF$

Super-heterodyne Receiver

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Introduction

- When you tune your radio from 98 MHz to 102 MHz, what actually changes inside?

- ✓ • Signal frequency changes
- ✓ • But internal processing frequency is mostly fixed

- Instead of adjusting everything, we convert all signals to one common frequency. That's the magic of Superheterodyne !

Core Idea of Superhetrodyne

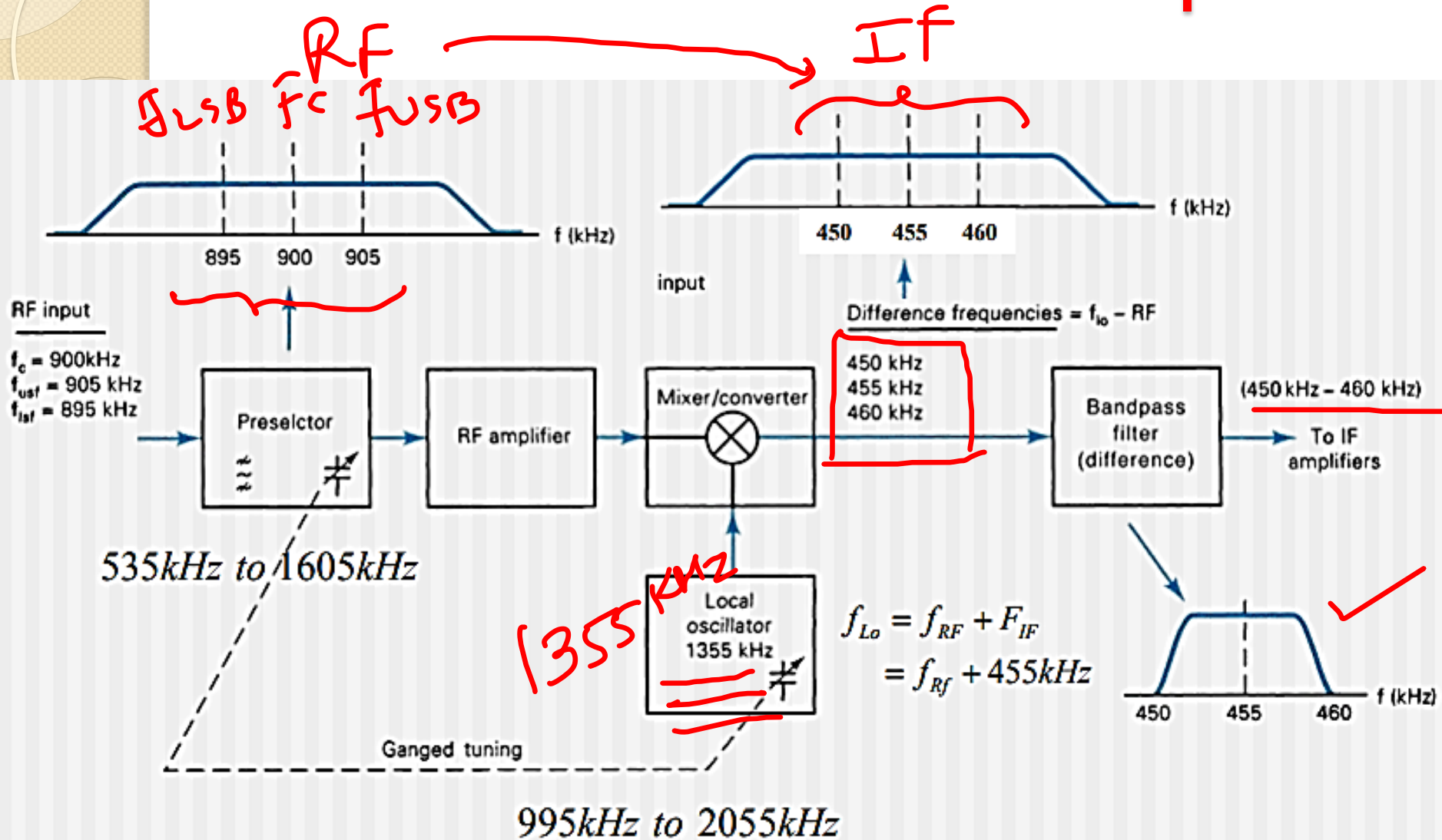
- Convert incoming signal ^{→ RF} to a fixed Intermediate Frequency (IF) → mixer & LO

$$✓ f_{IF} = |f_{RF} - f_{LO}|$$

- ✓ • f_{RF} is the signal frequency (incoming RF)
- ✓ • f_{LO} is the local oscillator frequency



RF-to-IF Conversion Example



Choice of IF: Factors

If the IF is too high ✓

- Poor Selectivity and Poor adjacent channel rejection.
- Tracking Difficulties.

If the IF is too low

- Image frequency rejection becomes poorer.
- Selectivity too sharp and cutting off sidebands
- Instability of oscillator will occur.

- The IF must not fall in the tuning range of the receiver.

Standard IF frequency

- AM broadcast (tuned to 540 to 1650 KHz) use IF in the range of 438-465 KHz. 455 KHz is more popular.

Am

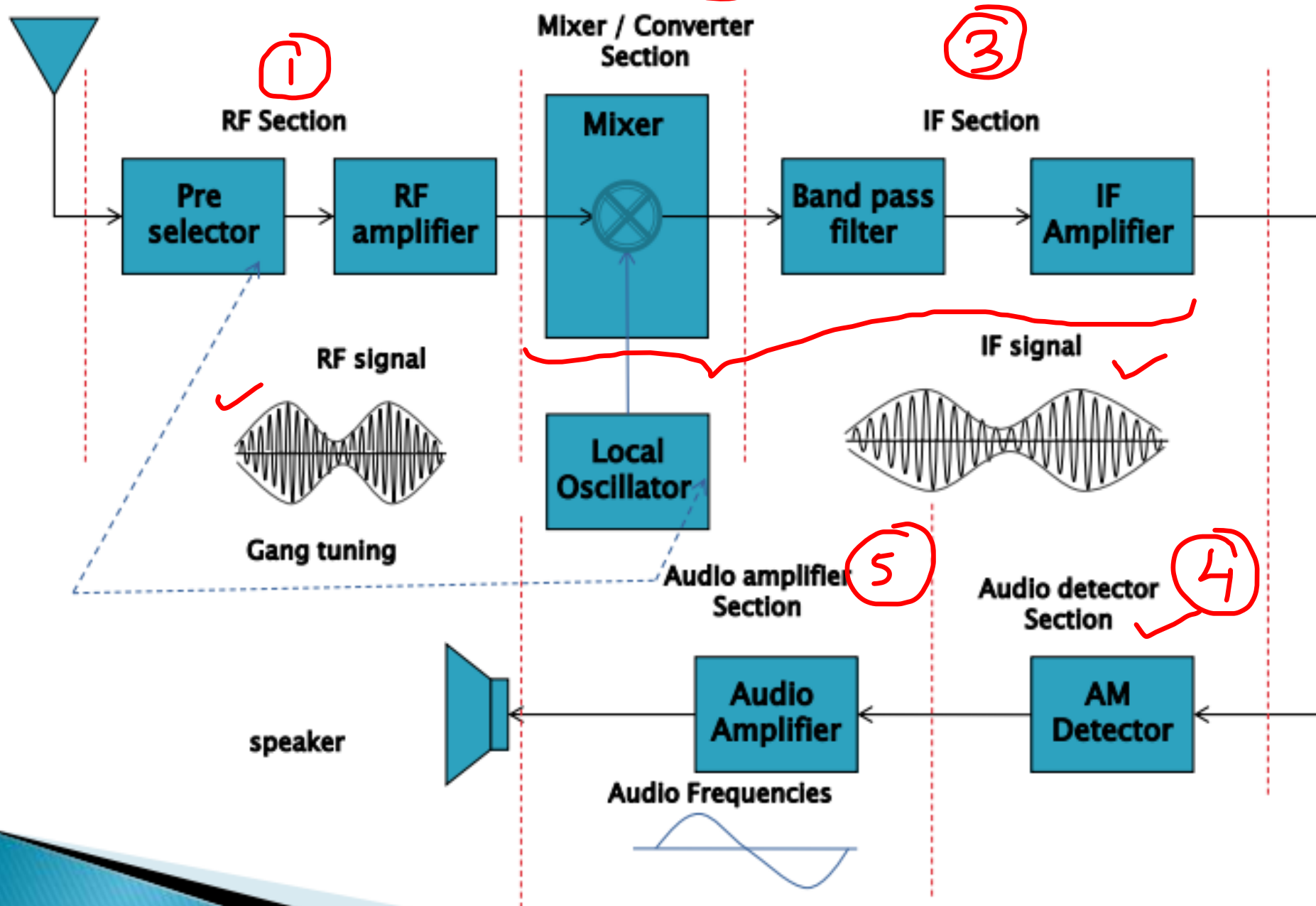
- AM, SSB uses 1.6 to 2.3 MHz.

- FM receiver using 88-108 MHz range uses IF of 10.7 MHz.

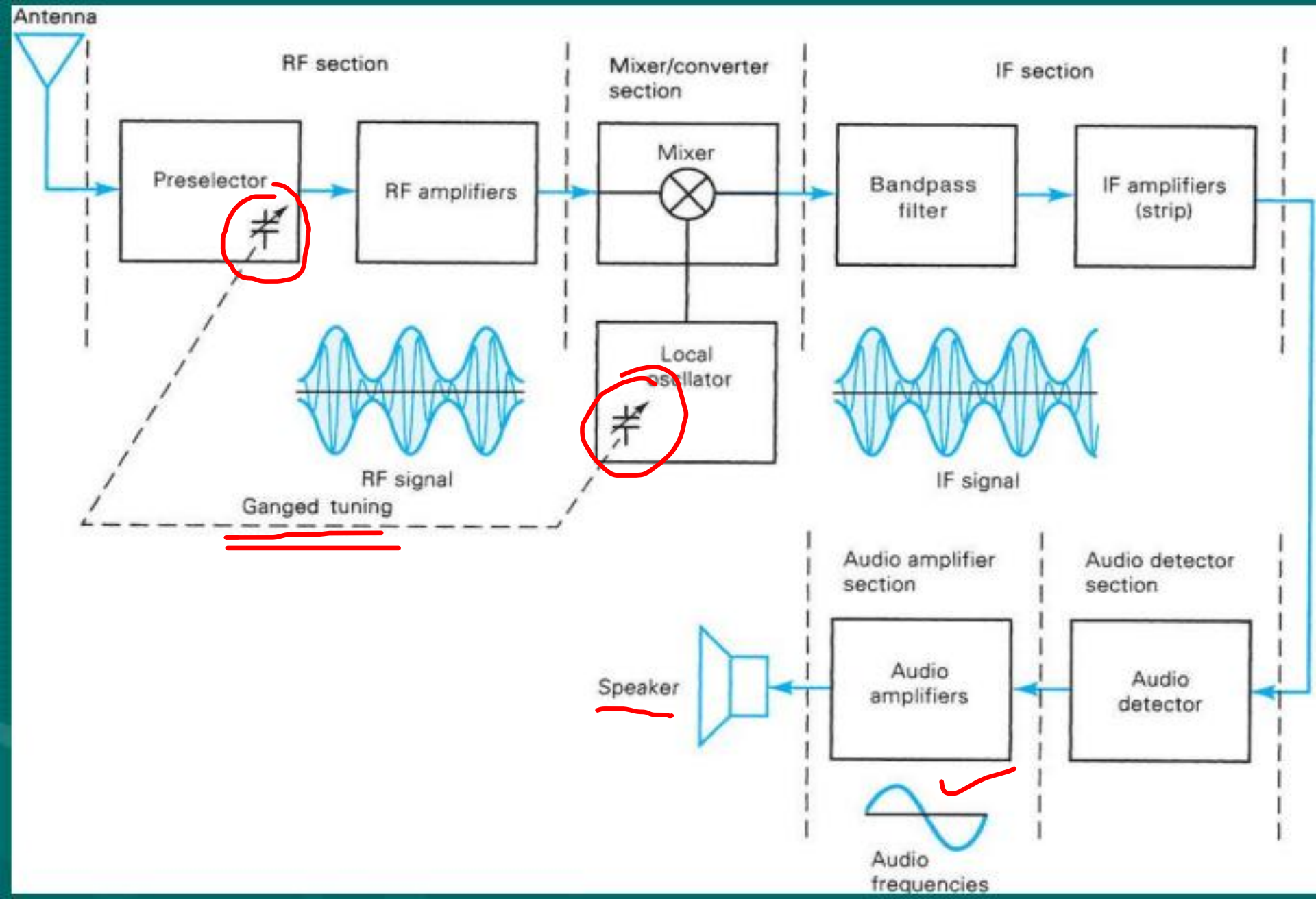
- TV receivers uses IF between 26 and 46 MHz.

- Microwave and RADAR receivers uses 70 MHz.

Superhetrodyne Receiver



Superhetrodyne Receiver (Repeat)





Superhetrodyne Receiver Sections

- **RF section**
- **Mixer/converter section**
- **IF section**
- **Audio detector section**
- **Audio amplifier section**



RF Section-Pre-selector Stage

Broad tuned band pass filter with adjustable frequency that is tuned to carrier frequency

- Provide initial band limiting to prevent specific unwanted radio frequency called image frequency from entering into receiver.
- Reduces the noise bandwidth of the receiver and provides the initial step toward reducing the over all receiver bandwidth to the minimum bandwidth required to pass the information signal.



RF Section-RF Amplifier

It determines the sensitivity of the receiver.

- RF amplifier is the first active device in the network it is the primary contributor to the noise. And it is the predominant factor in determining the noise figure.
- Receiver may have one or more RF amplifier depending on the desired sensitivity.

Due to RF Amplifier

- ▶ Greater gain and better sensitivity
- ▶ Improved image frequency rejection
- ▶ Better signal to noise ratio
- ▶ Better selectivity.



Mixer/Convertor Section

It consists of two components

- Mixer
- Local oscillator

Mixer stage :

- Heterodyning takes place in the mixer stage.
- Radio frequencies are down converted to intermediate frequency
- Carrier and sidebands are translated to high frequencies without effecting the envelope of message signal.

Frequency Conversion

- The difference between the RF and Local oscillator frequency is always constant IF.

$(LO > RF)$

High side injection



Local oscillator frequency is tuned above RF



$$f_{LO} = f_{RF} + f_{IF}$$

$(LO < RF)$

Low side injection

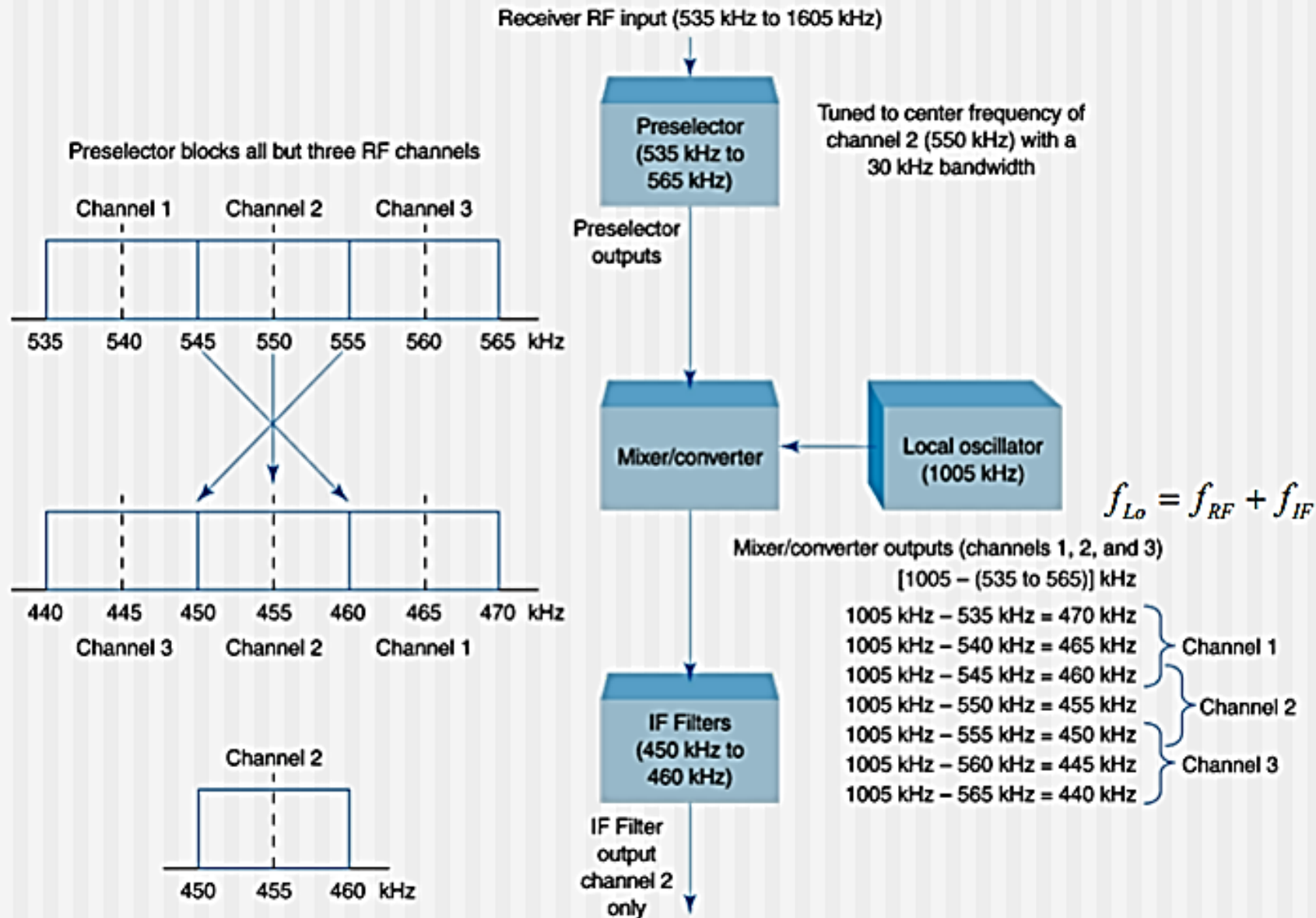


Local oscillator frequency is tuned below RF



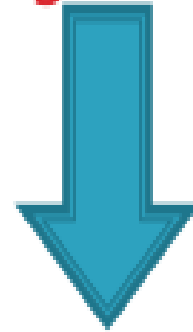
$$f_{LO} = f_{RF} - f_{IF}$$

RF to IF Conversion (Ex-I)



Gang Tuning

- ▶ The adjustment for the center frequency of the preselector and the adjustment for local oscillator are **gang tuned**.



The two adjustments are mechanically tied together and single adjustment will change the center frequency of the preselector and the local oscillator

Points to Remember

- Receiver must separate the desired signal from other signals and noise and then demodulate the signal.
- The most common receiver type is superheterodyne, which uses a mixer to transfer all incoming signals to a common IF.
- The RF amplifier is principal responsible for setting noise figure for receiver.
- Single-sideband receivers require a beat-frequency oscillator to reinsert the carrier and generally use product detector.